

Sai Rithwik Kukunuri

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EDUCATION

Arizona State University

Bachelor of Science in Computer Science (Barrett Honors College)

Minor: Data Science | Cumulative GPA: 4.00/4.00

Aug 2024 – May 2027

Tempe, Arizona

Relevant Coursework: Software Engineering, Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Information Assurance, Human-Computer Interaction, Machine Learning

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, C++, SQL
- **Frontend & Backend:** React, Next.js, CSS, HTML, FastAPI, Flask, Spring Boot, PostgreSQL, MySQL, MongoDB
- **AI & ML:** TensorFlow, PyTorch, Scikit-learn, NumPy, Groq, RAG pipelines, LangChain
- **Cloud & Tools:** AWS, Azure, Git, Docker, Figma, Linux

EXPERIENCE

Undergraduate Teaching Assistant

Arizona State University

May 2026 – Present

Tempe, Arizona

- Assist 35+ students in CSE 360 with software engineering concepts, project requirements, debugging, testing, and development tools.
- Support Java and JavaFX projects in IntelliJ by helping students improve code quality, software design, implementation, and debugging.
- Guide students on software engineering tools and practices, including JUnit testing, Astah UML diagrams, Jira, GitHub, and Scrum workflows.

IPv4 IPID Traffic Modeling and Network Security

Undergraduate Research Assistant, Biodesign Institute, ASU

Jan 2026 – Present

Tempe, Arizona

- Built a Python analysis pipeline with NumPy, SciPy, and Matplotlib to simulate bursty IPv4 network traffic and evaluate packet identifier behavior for network security analysis.
- Compared traditional Poisson traffic assumptions against MMPP-based bursty traffic models to identify security risks missed by prior analysis.
- Automated simulation, probability computation, and visualization workflows uncovering up to 63× higher vulnerability under real-world traffic patterns.

AI-Powered Question Answering for Medical Procedures

Undergraduate Researcher, TEAL Lab, ASU

Jan 2026 – Present

Tempe, Arizona

- Built a multimodal medical Q&A platform with Next.js, TypeScript, Flask, Qdrant, Groq APIs, and MedVidQA embeddings.
- Used Qdrant vector search to return top-5 relevant video segments, reducing latency by 30% and supporting 70 tokens/sec inference.
- Added text, image, voice, and video-frame inputs to return step-by-step answers with relevant video timestamps.

Arizona Technology Council: Data Integration and Insights

Project Lead

Sep 2025 – Dec 2025

Tempe, Arizona

- Led a 4-member team to build a full-stack data integration platform with React, TypeScript, Flask, REST APIs, and AWS.
- Automated data collection and transfer between Arizona Technology Council's two platforms, reducing reliance on manual workflows.
- Owned system architecture, sprint planning, and documentation, improving team coordination within a \$2,500 project budget.

PROJECTS

Bourse - AI Stock Research Platform | Next.js, FastAPI, MongoDB, FAISS, Groq, Finnhub API

[GitHub](#)

- Built an AI stock research platform to analyze 8,000+ U.S. stocks using natural-language queries and live market data.
- Developed a Groq-powered financial assistant using live prices, earnings, metrics, and news from Finnhub APIs.
- Implemented a FastAPI and MongoDB backend aggregating quotes, charts, financials, and news from 5+ endpoints in under 200ms.

Student Question and Answer System - Collaborative Learning Tool | Java, JavaFX, SQL, Git, Eclipse, Scrum

- Built a JavaFX collaborative Q&A application with SQL-backed CRUD features for posting questions, answering, and rating responses.
- Implemented role-based access control for students, instructors, admins, and staff to support moderation, feedback, and user management.
- Led a 5-member Scrum team through sprint planning, Git workflows, testing, and debugging to deliver core features on schedule.

LEADERSHIP

Google Developer Student Club

Graphic Designer

Sept 2025 – Present

Tempe, Arizona

- Designed and delivered visual content for developer workshops improving communication and engagement for a 2,000+ member community.
- Recognized as a 2026 Pitchfork Awards Finalist for contributions to student organization initiatives.

Software Developers Association

Marketing Officer

Jan 2025 – Present

Tempe, Arizona

- Promote workshops and developer events for a 3,000+ member community and design visual content to increase participation and outreach.
- Develop and maintain the organization's website features using React, TypeScript, and Flask improving usability and system functionality.